

Automated trading of VN30 Index stocks with boosting and expert weighting

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We consider a generic automated trading framework with an automatic model calibration of different versions of classical technical indicators based on their different parameters. The framework also employs an online learning process to be adaptive to market conditions. Eventually, a global layer is added to control risk.

Introduction

Statistical learning and technical analysis

There exists an expectation that a trader must know about although not necessarily use technical analysis indicators. However, an effective combination of the indicators and selection of winning parameters are always a hard decision, especially for a junior trader.

One naive approach is a “brute force” method: large number of alternatives are tested and the best combination is selected. This incurs expensive computation costs and it is not robust to changing market conditions as well. Within this context, statistical learning methods with feature selection capability are more intuitive and reliable.

Ensemble learning

When we try to predict the target variable using any machine learning technique, the main causes of difference in actual and predicted values are noise, variance, and bias. Ensemble helps to reduce these factors.

An ensemble is just a collection of predictors which come together to give a final prediction. The reason we use ensembles is that many different predictors trying to predict same target variable will perform a better job than any single predictor alone. Ensemble techniques are further classified into Bagging

and Boosting. Bagging is a simple ensembling technique in which we build many independent predictors/models/learners and combine them using some model averaging techniques. On contrary, boosting is an ensemble technique in which the predictors are not made independently, but sequentially. This technique employs the logic in which the subsequent predictors learn from the mistakes of the previous predictors. Because new predictors are learning from mistakes committed by previous predictors, it takes less time/iterations to reach close to actual predictions. [6]

Gradient boosting

Gradient boosting is a very popular boosting algorithm which produces a prediction model in the form of an ensemble of weak prediction models, typically decision trees. It builds the model in a stage-wise fashion like other boosting methods do, and it generalizes them by allowing optimization of an arbitrary differentiable loss function.

We consider a loss function as mean squared error:

$$Loss = MSE = \sum (y_i - y_i^p)^2 \quad (1)$$

where y_i = i -th target value and y_i^p = i -th prediction.

We aim our predictions such that our loss function (MSE) is minimum. By using gradient descent and updating our predictions based on a learning rate, we can find the values where MSE is minimum.

$$y_i^p = y_i^p + \alpha \times \frac{\partial \sum (y_i - y_i^p)^2}{\partial y_i^p} \quad (2)$$

which becomes,

$$y_i^p = y_i^p - 2\alpha \sum (y_i - y_i^p) \quad (3)$$

where α is the learning rate and $\sum (y_i - y_i^p)$ is the sum of residuals. [5]

Online learning and market adaptation

Statistical learning methods train models from a limit set of observations and produce trading rules which are not adaptive to frequent market changes. This is a reason that online learning methods become attractive for application of statistical learning methods in financial markets. Online learning methods evaluate performance/error of the previous rules and construct the new (combined) rule.

In this work, an algorithm of online learning with expert weighting was used is an extension applied to financial time series prediction [2].

Trading framework

This framework was structured into three layers including a machine learning algorithm, online learning, and risk management. The first layer consisted of experts generated by a machine learning algorithm, particularly gradient boosting method. This layer needs a learning algorithm with feature selection and importance ability so that it is able to select the best combination of trading rules and the rules' parameters. The online learning layer distributed different weights to these experts and outputs either buy, hold or sell decision. Eventually, the risk management layer validated trading signal strength and prevented the framework from short selling which is prohibited in Vietnamese stock markets. [2] [3]

Data preprocessing

Features

Technical indicators and technical trading rules were inputs to the learning model. They included simple moving average, exponential moving average, Bollinger bands, momentum, acceleration, rate of change, moving average convergence divergence (MACD), relative strength index (RSI), stochastic oscillators, Williams indicator, money flow index, Chaikin volatility, on balance volume, accumulation/distribution line, Chaikin oscillator, negative and positive volume index, and price-volume trend [2]. The rules and indicators were generated with different parameters to create a set of features $X = (x_1, x_2, \dots, x_n)$ and we let the machine learning algorithm to select the best combination of rules and parameters.

The figures were then normalized by the following method:

$$x'_i = \frac{x_i - \hat{\mu}_{x_i}}{\hat{\sigma}_{x_i}} \quad (4)$$

where $\hat{\mu}_{x_i}$ and $\hat{\sigma}_{x_i}$ are averages and standard deviations of the figures respectively.

Target classes

Our goal was to predict the trend of abnormal return on day t (AR_t) by using the above features on day $(t - 1)$ (X_{t-1}). The abnormal return was calculated as the difference between the return (R_t) and the benchmark (R_{bm_t}).

$$AR_t = R_t - R_{bm_t} \quad (5)$$

To refrain from excessive trades due to noises, we only classified statistically significant abnormal returns into either positive or negative by a parameter δ to reasonably reduce trading turnover and eventually save transaction costs. In particular, abnormal return AR_t that is greater than $\max(\hat{\mu}_{AR} + \delta\hat{\sigma}_{AR}, 0)$ was classified as positive and its target class is $y_t = +1$ signaling a buy decision. In contrast, abnormal return AR_t that is less than $\min(\hat{\mu}_{AR} + \delta\hat{\sigma}_{AR}, 0)$ is classified as negative and its target class is $y_t = -1$ signaling a sell decision. Otherwise, $y_t = 0$ signals a hold decision.

In this writing, VN30 Index was the benchmark. In addition, $\delta = 1$ was selected so that approximately one-third of observations will trigger a buy or sell decision at most if the stock returns are normally distributed. This reflects our preferred trading activeness.

Layer 1: Machine learning algorithm

The features were inputs to gradient boosting algorithm. Each trained model is an expert with its best combinations of signals. We aim to forecast either a positive, negative or insignificant abnormal returns relative to the benchmark, which is $y_t \in [-1, +1]$, where $+1$ signals a extreme positive abnormal return and -1 otherwise. In this sense, an objective of this learning algorithm was to outperform the benchmark.

Layer 2: Online learning and expert weighting

This second layer weighed and selected the best output out of the experts by producing a mutual recommendation based on expert weighting algorithm [2].

This algorithm produces a final outcome sequence $W = (W_1, W_2, \dots, W_t, \dots)$ from a sequence of experts $\psi = (\psi_1, \psi_2, \dots, \psi_E)$ where E is the number of experts.

The outcome of expert ψ_i at the time t is the prediction score S_t^i . In order to calculate the experts' weight, we need to calculate cumulative abnormal return of each expert $i(\psi_i)$ as

$$car_t^i = \sum_{s=t_i+1}^t \text{sign}(S_s^i) \times ar_s^i \quad (6)$$

where $t_1 = 0$, t_i is the time step at which ψ_i was added to the pool when $i > 1$, and ar_s^i is abnormal return for expert i at time s .

The weight of the first expert is

$$w_t^1 = \exp\left(\frac{C \times car_t^1}{\sqrt{t}}\right) \quad (7)$$

where C is an exogenous parameter.

The weight of expert ψ_i at time t where $t > t_i$ is

$$w_t^i = I_i \times \text{ramp}(t - t_i) \times \exp\left(\frac{C \times car_t^i}{\sqrt{t - t_i}}\right) \quad (8)$$

where $I_i = \sum_{j=1}^{i-1} w_{t_i}^j / (i - 1)$ is the initial weight assigned to the new expert $i(\psi_i)$ and is the average weight of the previous experts at the time that the new expert is added (t_i), $\text{ramp}(t - t_i) = \min([t - t_i] / [t_{i+1} - t_i], 1)$ is a function that brings the new expert gradually, and t_{i+1} is the time that the next expert is added.

The experts weight (W_t) is the result of weighting the answers of all the experts:

$$W_t = B_t - S_t \quad (9)$$

where the fraction of experts that suggest a buy or sell decision is $B_t = \sum_{i: S_t^i > 0} w_t^i / \sum_i w_t^i$ or $S_t = 1 - B_t$, respectively. Therefore, $W_t \in [-1, +1]$ is also a trading suggestion to take either a buy ($W_t > 0$), hold ($W_t = 0$) or a sell decision ($W_t < 0$).

The algorithm version is called the "Base" version. Additionally, we include two variations of the algorithm where we modify the denominator of the weight functions as follows.

1. "Simple time adjustment" version, where

$$w_t^i = I_i \times \text{ramp}(t - t_i) \times \exp\left(\frac{C \times car_t^i}{t - t_i}\right) \quad (10)$$

$$w_t^1 = \exp\left(\frac{C \times car_t^1}{t}\right) \quad (11)$$

2. "No time adjustment" version, where

$$w_t^i = I_i \times \text{ramp}(t - t_i) \times \exp(C \times car_t^i) \quad (12)$$

$$w_t^1 = \exp(C \times car_t^1) \quad (13)$$

These versions are different in terms of adjustment for elapsed time from the addition of experts by $(t - t_i)$ and t factors included in w_t^i and w_t^1 respectively. Compared to the Base version with its square-root adjustment $\sqrt{t - t_i}$ and $\sqrt{t}$, the Simple version has a faster adjustment with $(t - t_i)$ and t , while the No adjustment version excludes the time adjustment. The faster the adjustment is, the more the model is effective in predicting abnormal return trend in rapidly changing market conditions by allocating more weights on later experts.

Layer 3: Risk management and trading

This layer validated positions preferred by the period layers and allocated weights to member stocks in a portfolio based on their signal strengths.

Firstly, the risk management layer evaluated the strength of the trading signal W_t given by the layer 2. As W_t is normalized between a completely sell (-1) and completely buy (+1) position, the risk management system eliminated trading signals that are not above a non-zero threshold γ .

Secondly, we considered a portfolio with n stocks $S = (s_1, s_2, \dots, s_i, \dots, s_n)$ with composition as $\Phi = (\phi_1, \phi_2, \dots, \phi_i, \dots, \phi_n)$. We compute Φ_t based on the expert weights as followings.

For $W_t^i = 0$: $\phi_t^i = \phi_{t-1}^i$.

For $W_t^i > 0$:

$$\phi_t^i = (1 - \sum_{W_t^i=0} \phi_t^i) \times \frac{W_t^i}{\sum_{W_t^i>0} W_t^i} \quad (14)$$

For $W_t^i < 0$: $\phi_t^i = 0$.

Performance assessment

We used compounded annual growth rate (CAGR), Sharpe ratio (SR_t), Sterling ratio (St_t) and Informative ratio (IR_t) as performance criteria [2]. Return of the portfolio is calculated as below.

$$R_t = \Phi_t r_t \quad (15)$$

where Φ_t is the portfolio composition and r_t is a return vector of the member stocks at time t .

The Sharpe ratio is annualized and is calculated as the mean of monthly returns of divided by the standard deviation of the monthly returns:

$$SR_t = \frac{\mu(R_t)}{\sigma(R_t)} \quad (16)$$

The Sharpe ratio is possibly very unstable for small return variances and it cannot distinguish between

periods having large profits or losses. For this reason, Sterling ratio is added to detect clusters of profits and losses and it is calculated as the annualized difference between the return and the risk-free rate (R_f) divided by maximum drawdown ratio:

$$St_t = \frac{R_t - R_f}{D_t} \quad (17)$$

where D_t is maximum drawdown over a period ($t - t_0$): $D_t = \max(R_{t_x} - R_{t_y} | t_0 \leq t_x \leq t_y \leq t)$. In this study, we selected $R_f = 6\%$ [4] and a window ($t - t_0$) = 30 trading days. The Information ratio is also more robust than Sharpe ratio because it calculates excess risk-adjusted return from the benchmark. The Information ratio is annualized and is calculated as the mean of monthly abnormal return divided by the standard deviation of the abnormal return.

$$IR_t = \frac{\mu(AR_t)}{\sigma(AR_t)} \quad (18)$$

where $AR_t = R_t - R_{bm_t}$.

Experiments and empirical results

Experiment design

We selected 22 out of 30 stocks with daily data from January 2013 and December 2017 in VN30 Index at the end of 2017. The thirty component stocks of VN30 Index are outlined in the appendix. In a discretionary manner, we left out ROS and SAB. The reason was because their transparency of their trading activities was subjectively questionable to us so their application of technical indicators would be not credible. In addition, BID, NVL and MWG were listed after January 2013 and they were also not included in our study. Last but not least, SSI and NT2 were also left out due to incomplete data. For every stock we had a series of close, open, high, low prices and volumes. We obtained these price series from *cophieu68.com* website [1]. The time series were distributed in the following way.

- The training set included two years. This data set includes data from January 2013 to December 2014. After discarding 37 lagged values, we had 460 observations for each stock, including data from March 2013 to December 2014. We aggregated in one set the observations of all the stocks. In this sense, our training set had 10120 observations.
- The validation set is based on the next 200 trading days (4400 observations). This dataset

includes data from January 2015 to October 2015.

- The test set had the next 549 trading days (12078 observations). This dataset includes data from November 2015 to December 2017.

Optimization and training

We trained an expert on the training set and used it as the mutual initial expert for all stocks because it was more likely to predict diverse patterns of the stocks. Then, we ran our algorithm using a moving window of 50 trading days and we tested the existent experts every day with the additional price information. The expert weighting algorithm re-weighted the participation of each expert according to individual performance and time elapsed since the initial training. Eventually, the risk management and trading disposed of weak signals predicted by the experts and computed portfolio composition.

We ran the expert weighting algorithm using its three versions and the following values of the parameters C : 0.05, 1, 10, and 50; γ : 0, 0.25, 0.5 and 0.75. When C is 0, this is equivalent to averaging all the experts. In validation stage to select optimal parameters, we randomly generated with replacement 3 portfolios consisting of 10 member stocks. The average of the portfolios' performance criteria described in the performance assessment section was validated to select optimal parameters.

Optimization results are presented in Figure 1, Figure 2, Figure 3 and Figure 4. The larger C was, the better its resulting CAGR, Sharpe ratio, Sterling ratio, and Informative ratio became. Larger values of C place higher emphasis on experts' cumulative abnormal returns and the elapsed-time factor which is applicable for Base and Simple models. However, enormous values of C will disregard prematurity of experts (i.e. via $\text{ramp}(t - t_i)$) and their relevance to earlier experts (i.e. via their initial weights I_i).

In addition, $\gamma = 0$ was the optimal threshold for all of the models. Hence, $C = 50$ and $C = 0$ (Average Weight model), together with $\gamma = 0$, were presented when we tested out-of-sample data.

Moreover, No-adjustment model outperformed the others and its result was expected to be the most promising for out-of-sample data.

Results

Testing results of CAGRs, Sharpe ratio, Sterling ratio and Informative ratio are presented in Table

1, Table 2, Table 3 and Table 4. All of the models outperformed VN30 Index in the criteria in the out-of-sample data. In addition, as expected, No adjustment model had the best performance as it did in the in-sample data. Its equity curve generally tripled annually and its Sharpe ratio, Sterling ratio and Informative ratios were about 4. As a runner-up, the base model had its equity curve nearly doubled annually and its ratios were around 2. Next, the average weight and simple models had similar performances with their CAGRs about 60 percent and their ratios ranging from 1 to 1.4.

The above observation confirmed the discussed roles of experts' cumulative performance and elapsed-time factor in the optimization. The Vietnamese stock markets probably did not change rapidly in a way that made early existent experts ineffective. Also, predicting power of the initial expert might be significant due to its large training set so its weight should not have been lessened by time factor as by the base and simple models.

Figure 5 and Figure 6 demonstrate performance of the models and VN30 Index in the 200 in-sample and the next 549 out-of-sample trading days respectively. The no adjustment and base models substantially outperformed VN30 Index in both periods, while the average weight and simple surpassed VN30 Index from the beginning of the out-of-sample data.

However, when different transaction costs (tc) were taken into account, this was sequentially no longer the case. In this study, all transactions were assumed to be executed at close prices. Returns after transaction costs are calculated as the following.

$$R_{tc_t} = R_t - tc \times |\Phi_t - \Phi_{t-1}|r_t \quad (19)$$

For $tc = 0.01\%$ in Figure 7, equity curves of the average weight and simple models ended up at the same level of VN30 Index (i.e. about 1.5). For $tc = 0.02\%$ in Figure 8, only the no adjustment model still outperformed the index; however, it was caught up by the index with $tc = 0.03\%$ in Figure 9.

Conclusion

In this study, the no adjustment and base models significantly outperformed VN30 Index in recommending trading decisions without transaction costs November 2015 to December 2017 with a layered structure including gradient boosting method, online learning, and risk management. A portfolio consisted of 22 out of 30 component stocks in VN30 Index and

it was rebalanced every trading day. The predominant performance of the no adjustment model was still present for transaction costs equal to less than 0.02\$ per transaction.

Importantly, the assumption that the portfolio could be rebalanced every trading day limits application of this framework into current Vietnamese market conditions where the settlement takes three days to complete. Hence, it would substantially increase practical values of this framework if a longer rebalancing cycle is modeled.

In addition, C larger than 50 should be also experimented to find an optimal weight of exponential of the experts' performance indicator. Last but not least, other boosting methods, namely, AdaBoost and XGBoost are promising to apply in this structure beside gradient boosting.

References

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Figures and Tables

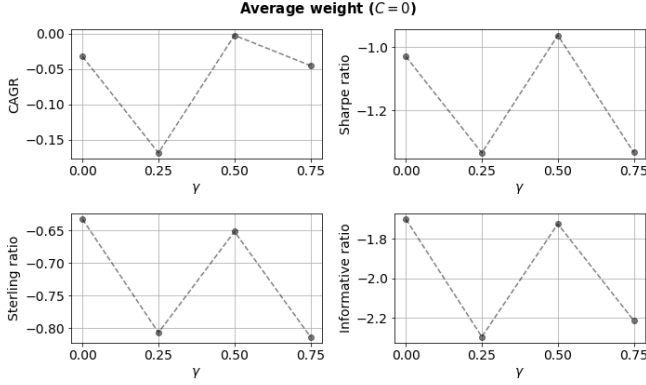


Figure 1: Compounded annual growth rate (CAGR), Sharpe ratio, Sterling ratio, and Informative ratio by a positive γ with no transaction costs of Average Weight model ($C = 0$). The risk management system eliminated trading signals that are not above threshold γ . The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. $\gamma = 0$ and $\gamma = 0.5$ were optimal choices for this model because they yielded the best results of the criteria.

Table 1: Compounded annual growth rate (CAGR) by model with no transaction costs, $C = 50$ and $\gamma = 0$ in 200 in-sample trading days (January 2015 to October 2015) and 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure.

Model	In-sample	Out-of-sample
Avg weight	-9.35%	61.11%
Base	48.96%	111.79%
Simple	0.44%	64.70%
No adjustment	121.94%	192.57%
VN30	2.16%	23.85%

Table 2: Sharpe ratio by model with no transaction costs, $C = 50$ and $\gamma = 0$ in 200 in-sample trading days (January 2015 to October 2015) and 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure.

Model	In-sample	Out-of-sample
Avg weight	-1.04	1.33
Base	0.33	2.02
Simple	-0.79	1.40
No adjustment	0.90	3.91
VN30	-0.62	1.08

Table 3: Sterling ratio by model with no transaction costs, $C = 50$ and $\gamma = 0$ in 200 in-sample trading days (January 2015 to October 2015) and 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure.

Model	In-sample	Out-of-sample
Avg weight	-0.59	1.04
Base	0.28	1.95
Simple	-0.48	1.12
No adjustment	1.08	4.48
VN30	-0.36	0.91

Table 4: Informative ratio by model with no transaction costs, $C = 50$ and $\gamma = 0$ in 200 in-sample trading days (January 2015 to October 2015) and 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure.

Model	In-sample	Out-of-sample
Avg weight	-1.23	0.99
Base	0.68	1.86
Simple	-0.82	1.08
No adjustment	1.27	4.09
VN30	1.09	-0.41

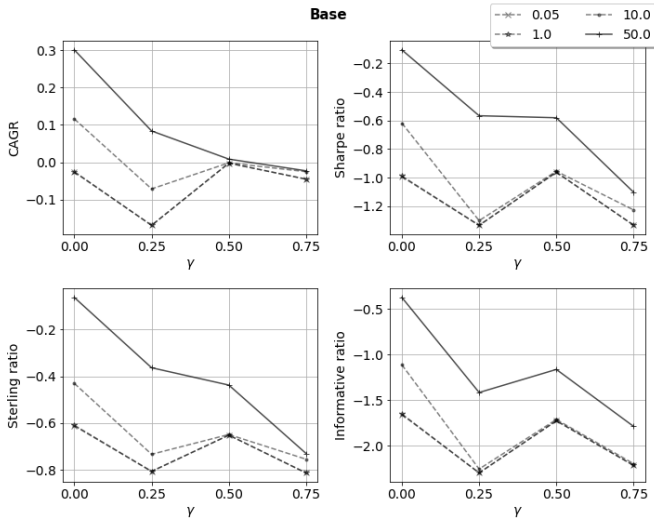


Figure 2: Compounded annual growth rate (CAGR), Sharpe ratio, Sterling ratio, and Informative ratio by a positive C and a positive γ with no transaction costs of Base model. The risk management system eliminated trading signals that are not above threshold γ . The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. $C = 50$ and $\gamma = 0$ was the optimal choice for this model because they yielded the best results of the criteria.

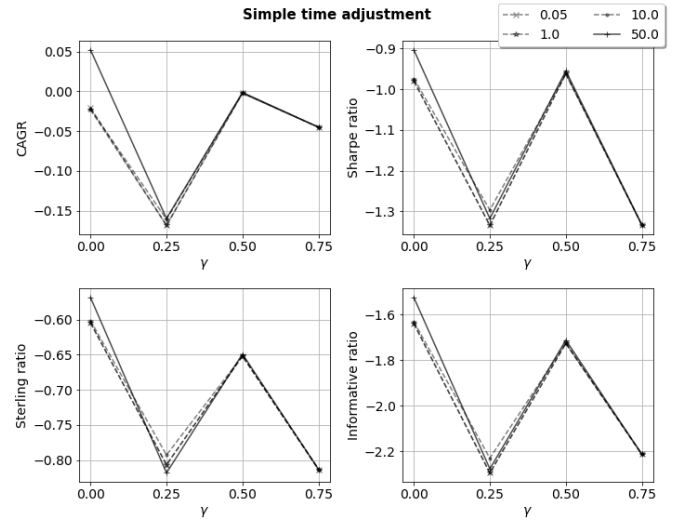


Figure 3: Compounded annual growth rate (CAGR), Sharpe ratio, Sterling ratio, and Informative ratio by a positive C and a positive γ with no transaction costs of Simple time adjustment model. The risk management system eliminated trading signals that are not above threshold γ . The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. $C = 50$, and $\gamma = 0$ or $\gamma = 0.5$ were the optimal choices for this model because they yielded the best results of the criteria.

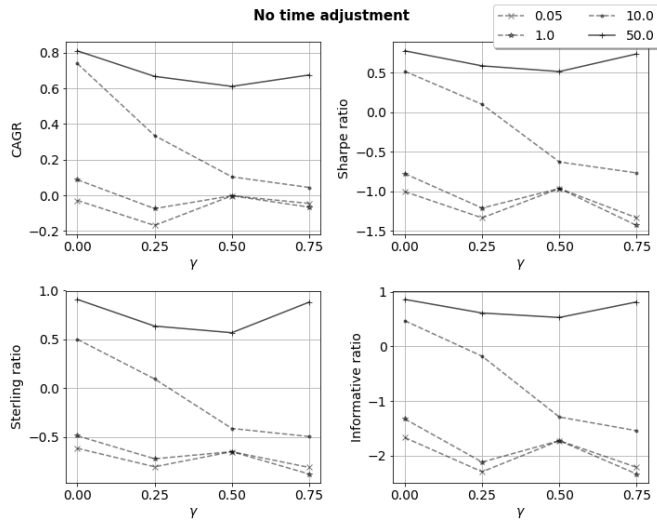


Figure 4: Compounded annual growth rate (CAGR), Sharpe ratio, Sterling ratio, and Informative ratio by a positive C and a positive γ with no transaction costs of No time adjustment model. The risk management system eliminated trading signals that are not above threshold γ . The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. $C = 50$, and $\gamma = 0$ or $\gamma = 0.75$ were the optimal choices for this model because they yielded the best results of the criteria.

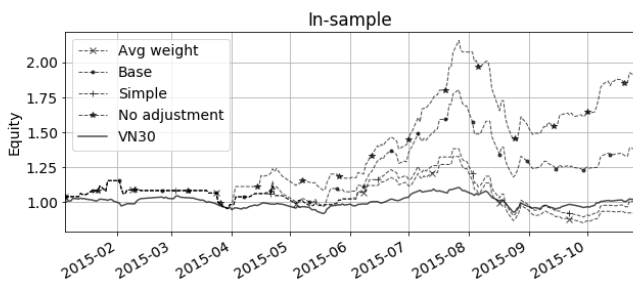


Figure 5: Cumulative equity curves with no transaction costs by model, with $C = 50$ and $\gamma = 0$, compared to VN30 Index in 200 in-sample trading days (January 2015 to October 2015). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. In the first 50 trading days, the models had only the initial expert so they had identical performance. The reason was because the rolling period to train additional expert was 50 days.

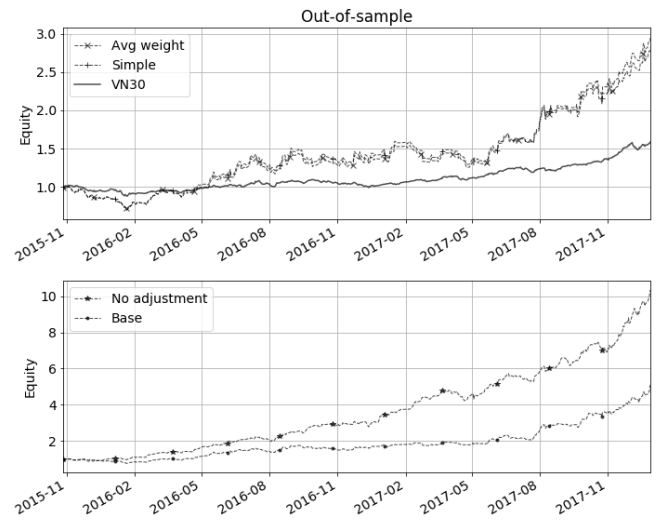


Figure 6: Cumulative equity curves with no transaction costs by model, with $C = 50$ and $\gamma = 0$, compared to VN30 Index in 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. No adjustment and Base models significantly outperformed the others so two separate charts are presented for clear visualization.

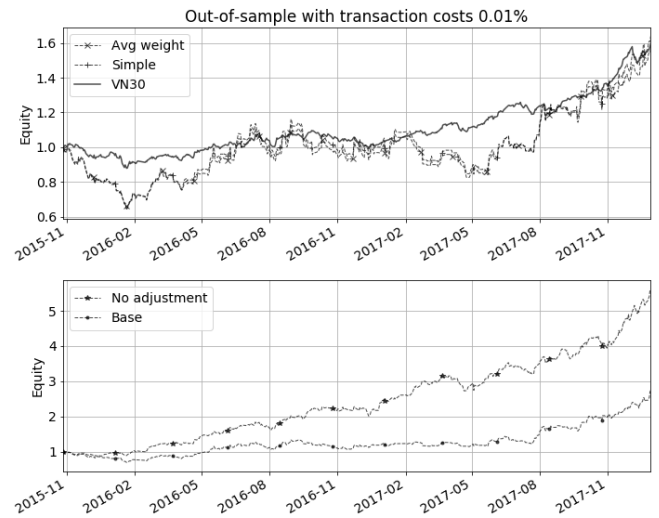


Figure 7: Cumulative equity curves with 0.01% per transaction by model, with $C = 50$ and $\gamma = 0$, compared to VN30 Index in 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure. No adjustment and Base models significantly outperformed the others so two separate charts are presented for clear visualization.

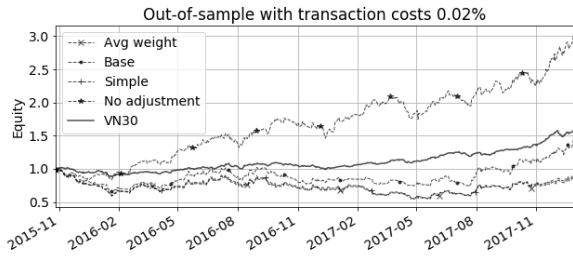


Figure 8: Cumulative equity curves with 0.02% per transaction by model, with $C = 50$ and $\gamma = 0$, compared to VN30 Index in 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure.

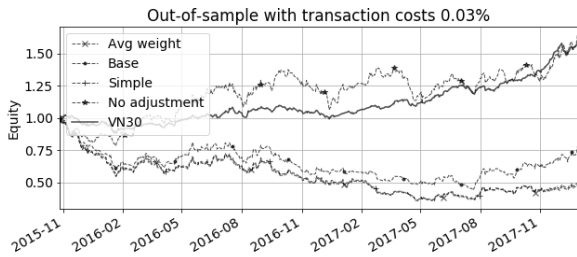


Figure 9: Cumulative equity curves with 0.02% per transaction by model, with $C = 50$ and $\gamma = 0$, compared to VN30 Index in 549 out-of-sample trading days (November 2015 to December 2017). The portfolio consisted of 22 out of 30 stocks in VN30 Index and it was rebalanced every trading day based on predicted signals of the layered trading structure.

Appendix A: Component stocks in VN30 Index

No	Securities symbol	Company name
1	BID	Joint Stock Commercial Bank for Investment and Development of Vietnam
2	BMP	Binh Minh Plastics Joint Stock Company
3	BVH	Bao Viet Holdings
4	CII	Hochiminh City Infrastructure Investment Joint Stock Company
5	CTD	Coteccons Construction Joint Stock Company
6	CTG	Viet Nam Joint Stock Commercial Bank For Industry And Trade
7	DHG	DHG Pharmaceutical Joint Stock Company
8	DPM	Petrovietnam Fertilizer And Chemicals Corporation
9	FPT	FPT Corporation
10	GAS	PetroVietnam Gas Joint Stock Corporation
11	GMD	Gemadept Corporation
12	HPG	Hoa Phat Group Joint Stock Company
13	HSG	Hoa Sen Group
14	KBC	Kinh Bac City Development Holding Corporation
15	KDC	KIDO Group Corporation
16	MBB	Military Commercial Joint Stock Bank
17	MSN	MaSan Group Corporation
18	MWG	Mobile World Investment Corporation
19	NT2	PetroVietnam Power Nhon Trach 2 Joint Stock Company
20	NVL	No Va Land Investment Group Corporation
21	PVD	Petrovietnam Drilling & Well Service Corporation
22	REE	Refrigeration Electrical Engineering Corporation
23	ROS	FLC FAROS Construction Joint Stock Company
24	SAB	Saigon Beer – Alcohol – Beverage Corporation
25	SBT	Thanh Thanh Cong - Bien Hoa Joint Stoc Company
26	SSI	Saigon Securities Incorporation
27	STB	Sai Gon Thuong Tin Commercial Joint Stock Bank
28	VCB	Joint Stock Commercial Bank For Foreign Trade Of Viet Nam
29	VIC	VINGROUP Joint Stock Company
30	VNM	Viet Nam Dairy Products Joint Stock Company

Appendix B: Investment signals for automated trading system

Investment signals for automated trading system		
Variable price indicators	Description	Calculation detail
$EMA_t^c(\lambda)$	Exponential moving average of a time series p^c	$EMA_t(p^c, \lambda)$, where $\lambda = 0.9, 0.84$ and 0.78
$rule1_t - rule4_t$	Exponential moving average to price ($p_t^c, p_t^{med}, p_t^{typ}, p_t^{wc}$)	$\frac{EMA_t^c(\lambda)}{p^c}, \frac{EMA_t^{med}(\lambda)}{p^{med}}, \frac{EMA_t^{typ}(\lambda)}{p^{typ}}, \frac{EMA_t^{wc}(\lambda)}{p^{wc}}$, where $n = 0.9, 0.84$ and 0.78
$SMA_t^c(n)$	Simple moving average of the last n observations of a time series p^c	$SMA_t(p^c, n)$ where $n = 10, 16$ and 22
$rule5_t$	Simple moving average to p_t^c	$\frac{SMA_t^c(n)}{p^c}$, where $n = 10, 16$ and 22
Bollinger bands	Using moving average or the median band ($Boll_t^m(n)$) as the reference point, the upper and lower Bollinger bands ($Boll_t^u(n)$ and $Boll_t^d(n)$), respectively) are calculated as a function of s standard deviation. When price crosses above (below) the upper (lower) Bollinger band, it is a sign that the market is overbought(oversold). Technical analysts typically calculate Bollinger bands using 20 days for the moving average and 2 standard deviations	$Boll_t^m(n) = SMA_t^c(n)$
$Boll_t^u(n)$	Upper Bollinger band	$Boll_t^m(n) + s\sigma_t^2(n)$, where $s = 2, n = 20, 26$ and 32
$Boll_t^d(n)$	Lower Bollinger band	$Boll_t^m(n) - s\sigma_t^2(n)$, where $s = 2, n = 20, 26$ and 32
Momentum and oscillation indicators		

$MOM_t(n)$	Momentum: price (p_t^c) change in the last n periods. When it crosses above(below) zero, it indicates that trend is up(down). The default value of n is 12	$p_t^c - p_{t-n}^c$, where n = 12, 18 and 24
$MomEMA_t(n, \lambda)$	Momentum to $EMA_t(MOM_t(n), \lambda)$	$\frac{MOM_t(n)}{EMA_t(MOM_t(n), \lambda)}$, where n =12, 18, 24 and $\lambda = 0.75$
$rule7_t$	Momentum trading rule	$\begin{cases} Buy & \text{if } MOM_{t-1}(n) \leq EMA_t(MOM_t(n), \lambda) \\ & \text{and } MOM_t(n) > EMA_t(MOM_t(n), \lambda) \\ Sell & \text{if } MOM_{t-1}(n) \geq EMA_t(MOM_t(n), \lambda) \\ & \text{and } MOM_t(n) < EMA_t(MOM_t(n), \lambda) \\ Hold & \text{otherwise} \end{cases}$
$ROC_t(n)$	Rate of change: rate of change of p_t^c . Technical analysts recommend using 10 periods to calculate this indicator	$\frac{p_t^c - p_{t-n}^c}{p_{t-n}^c} \times 100$, where n = 10, 16 and 22
$rule9_t$	ROC trading rule	$\begin{cases} Buy & \text{if } ROC_{t-1}(n) \leq 0 \text{ and } ROC_T(n) > 0 \\ Sell & \text{if } ROC_{t-1}(n) \geq 0 \text{ and } ROC_T(n) < 0 \\ Hold & \text{otherwise} \end{cases}$
$MACD_t(s, f)$	Moving average convergence divergence: difference between two moving averages of slow and fast periods (s,f). $MACD_t(s, f)$ is regularly calculated using 26(s) and 12(f) periods	$EMA_t(p^c, s) - EMA(p^c, f)$, where f = 12 and s = 18, 24 and 30
$MACD_t(s, f, n)$	MACD signal line: moving average of $MACD_t(s, f)$ of past n periods. A buy(sell) signal is generated when the $MACD_t(s, f)$ crosses above(below) the signal line or a threshold	$EMA_t(MACD(s, f), n)$, where f= 12, n= 9 and s= 18, 24, 30
$rule10_t$	MACD trading rule	$\begin{cases} Buy & \text{if } MACD_{t-1}(s, f) \leq MACD_{t-1}(s, f, n) \\ & \text{and } MACD_t(s, f) > MACD_t(s, f, n) \\ Sell & \text{if } MACD_{t-1}(s, f) \geq MACD_{t-1}(s, f, n) \\ & \text{and } MACD_t(s, f) < MACD_t(s, f, n) \\ Hold & \text{otherwise} \end{cases}$

$RSI_t(n)$

Relative strength index: compares the days that stock prices finish up against those periods that stock prices finish down. Technical analysts calculate this indicator using 9, 14 or 25 periods. A buy signal is when $RSI_t(n)$ crosses below a lower band of 30 (oversold), and a sell signal when $RSI_t(n)$ crosses below a lower band of 30(oversold), and a sell signal when $RSI_t(n)$ crosses above an upper band of 70 (overbought)

$rule11_t$

RSI trading rule

Stochastic oscillator

Compare close price to a price in a given period to establish if market is moving to higher or lower levels or is just in the middle. The oscillator indicators are as follows

$FAST\%K_t(n)$

Percentage measure of the last close price in relation to the highest high and lowest low of the last n periods (true range).

Vector with low prices of last n period

Vector with high prices of last n period

$FAST\%D_t(n)$

Moving average of $FAST\%K_t(n)$

$SLOW\%K_t(n)$

Identically calculated to $FAST\%D_t(n)$ using a 3-period moving of $FAST\%K_t(n)$

$SLOW\%D_t(n)$

Moving average of $SLOW\%K_t(n)$

$$100 - \frac{100}{1 + \frac{SMA_t(p_n^{up}, n1)}{SMA_t(p_n^{dn}, n1)}}, \text{ where } n1 = 8, 14 \text{ and } 20$$

and n is the length of the time series

$$p_t^{up} = \begin{cases} p_t^c & \text{if } p_t^c < p_{t-1}^c \\ empty & \text{otherwise} \end{cases}$$

$$p_t^{dn} = \begin{cases} p_t^c & \text{if } p_t^c < p_{t-1}^c \\ empty & \text{otherwise} \end{cases}$$

$$p_n^{up} = (p_{t-n}^{up}, p_{t-n+1}^{up}, p_{t-n+2}^{up}, \dots, p_t^{up})$$

$$p_n^{dn} = (p_{t-n}^{dn}, p_{t-n+1}^{dn}, p_{t-n+2}^{dn}, \dots, p_t^{dn})$$

$$\begin{cases} Buy & \text{if } RSI_{t-1}(n) \geq 30 \text{ and } RSI_t(n) < 70 \\ Sell & \text{if } RSI_{t-1}(n) \leq 30 \text{ and } RSI_t(n) > 70 \\ Hold & \text{otherwise} \end{cases}$$

$$\frac{p_t^{uc} - \min((P_n^l))}{\max(P_n^h) - \min(P_n^l)}, \text{ where } n=12, 18 \text{ and } 24$$

$$(P_n^l) = (P_{t-n}^l, P_{t-n+1}^l, P_{t-n+2}^l, \dots, P_t^l)$$

$$textbf{f}(P_n^h) = (P_{t-n}^h, P_{t-n+1}^h, P_{t-n+2}^h, \dots, P_t^h)$$

$$SMA_t(FAST\%K_t(n), 3)$$

$$SMA_t(FAST\%K_t(n), 3)$$

$$SMA_t(SLOW\%K_t(n), 3)$$

$rule12_t$	Fast stochastic trading rule	$\begin{cases} Buy & \text{if } FAST\%K_{t-1}(n) \leq FAST\%D_t(n) \\ & \text{and } FAST\%K_t(n) > FAST\%D_t(n) \\ Sell & \text{if } FAST\%K_{t-1}(n) \geq FAST\%D_t(n) \\ & \text{and } FAST\%K_t(n) < FAST\%D_t(n) \\ Hold & \text{otherwise} \end{cases}$
$rule13_t$	Slow stochastic trading rule	$\begin{cases} Buy & \text{if } SLOW\%K_{t-1}(n) \leq SLOW\%D_t(n) \\ & \text{and } SLOW\%K_t(n) > SLOW\%D_t(n) \\ Sell & \text{if } SLOW\%K_{t-1}(n) \geq SLOW\%D_t(n) \\ & \text{and } SLOW\%K_t(n) < SLOW\%D_t(n) \\ Hold & \text{otherwise} \end{cases}$
$slowKslowD_t(n)$	$SLOW\%K_t(n)$ to $SLOW\%D_t(3)$	$\frac{SLOW\%K_t(n)}{SLOW\%D_t(3)}$
$fastKfastD_t(n)$	$FAST\%K_t(n)$ to $FAST\%D_t(3)$	$\frac{FAST\%K_t(n)}{FAST\%D_t(3)}$
$WILL_t(n)$	William indicator: the calculation is similar to the stochastic oscillator with a scale from 0 to -100. It tries to capture moments when the market is overbought(0-20) or oversold(-80-100)	$\frac{\max(\mathbf{P}_n^h) - p_t^{uc}}{\max(\mathbf{P}_n^h) - \min(\mathbf{P}_n^l)}$
$rule14_t$	Williams trading rule	$\begin{cases} Buy & \text{if } WILL_{t-1}(n) \geq -20 \text{ and } WILL_t(n) < -80 \\ Sell & \text{if } WILL_{t-1}(n) \leq -20 \text{ and } WILL_t(n) > -80 \\ Hold & \text{otherwise} \end{cases}$
$MFI_t(n)$	Money flow index: measures the strength of money flow (MF_t) in and out of a stock. At difference of the $RSI_t(n)$ which is calculated using stock prices, $MFI_t(n)$ is calculated using volume	$100 - \frac{100}{1 + \frac{PMF_t(n)}{NMF_t(n)}}$ Where: $n = 14$ $MF_t = P_t^{typ} \times VOL_t$ $PMF_t(n) = \mathbf{SMA}_t(MF_t, n)$ when $MF_t > 0$ $NMF_t(n) = \mathbf{SMA}_t(MF_t, n)$ when $MF_t < 0$ VOL_t is volume of day t $PMF_t(n)$ is positive money flow $NMF_t(n)$ is negative money flow
$rule15_t$	Money flow index trading rule. When $MFI_t(n)$ crosses above (below) 70 (30), this is a sign that the market is overbought (oversold)	$\begin{cases} Buy & \text{if } MFI_{t-1}(n) \geq 30 \text{ and } MFI_t(n) < 70 \\ Sell & \text{if } MFI_{t-1}(n) \leq 30 \text{ and } MFI_t(n) > 70 \\ Hold & \text{otherwise} \end{cases}$